Circuit Theory Course Project

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Project Introduction

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part one

Problem 4: Transfer Functions

Using the internal node voltage expressions, the following transfer functions are defined:

$$H_{HP}(s) = \frac{V_1}{V_i}, \qquad H_{BP}(s) = \frac{V_2}{V_i}, \qquad H_{LP}(s) = \frac{V_3}{V_i} \tag{1}$$

Rewrite each transfer function in the standard second-order form given by:

$$H(s) = \frac{N(s)}{s^2 + \frac{\omega_n}{Q}s + \omega_n^2} \tag{2}$$

Clearly identify the numerator polynomial $N(s)$, the natural frequency ω_n, and the quality factor Q for each case.

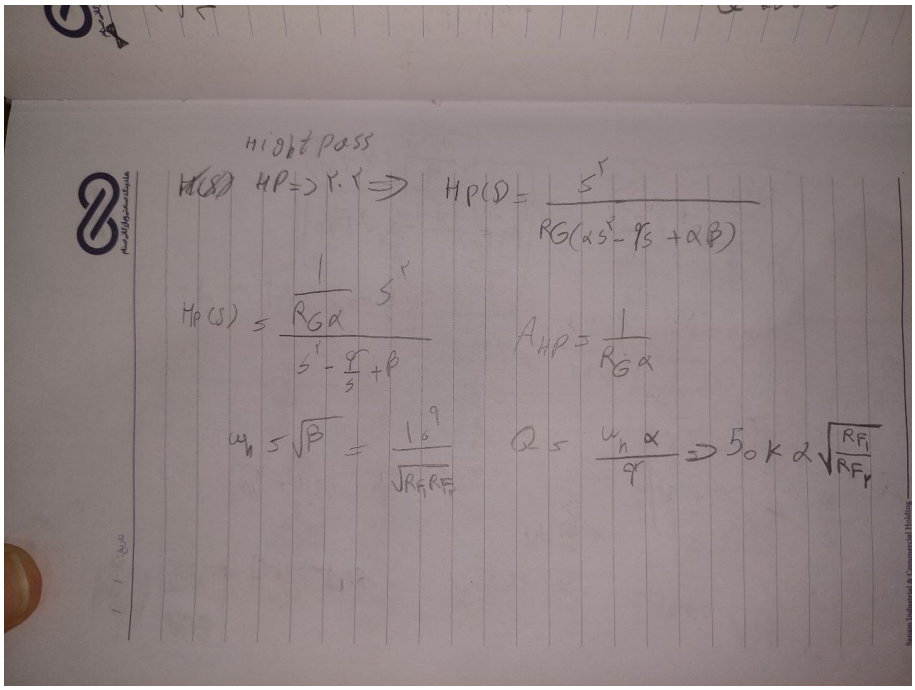


Figure 1: Handwritten solution for Part I

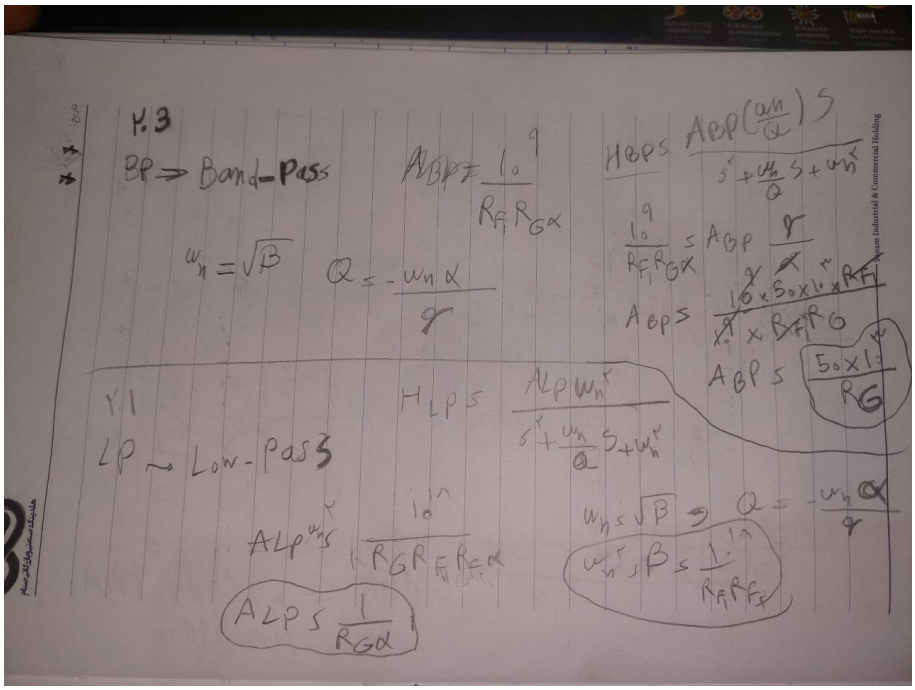


Figure 2: Handwritten solution for Part I

Problem 5: Filter Design

Based on your previous calculations, derive the expressions for the quality factor Q and the cutoff frequency f_c in terms of the circuit parameters. Then, design suitable values for the resistors R_G, R_{F1}, R_{F2}, and R_Q such that the following specifications are satisfied:

$$f_c = 20\ kH, \qquad Q = 0.75 \tag{3}$$

Provide clear and detailed calculations.

Assume the following simplifying conditions:

$$R_G = R_Q, \qquad R_{F1} = R_{F2} \tag{4}$$

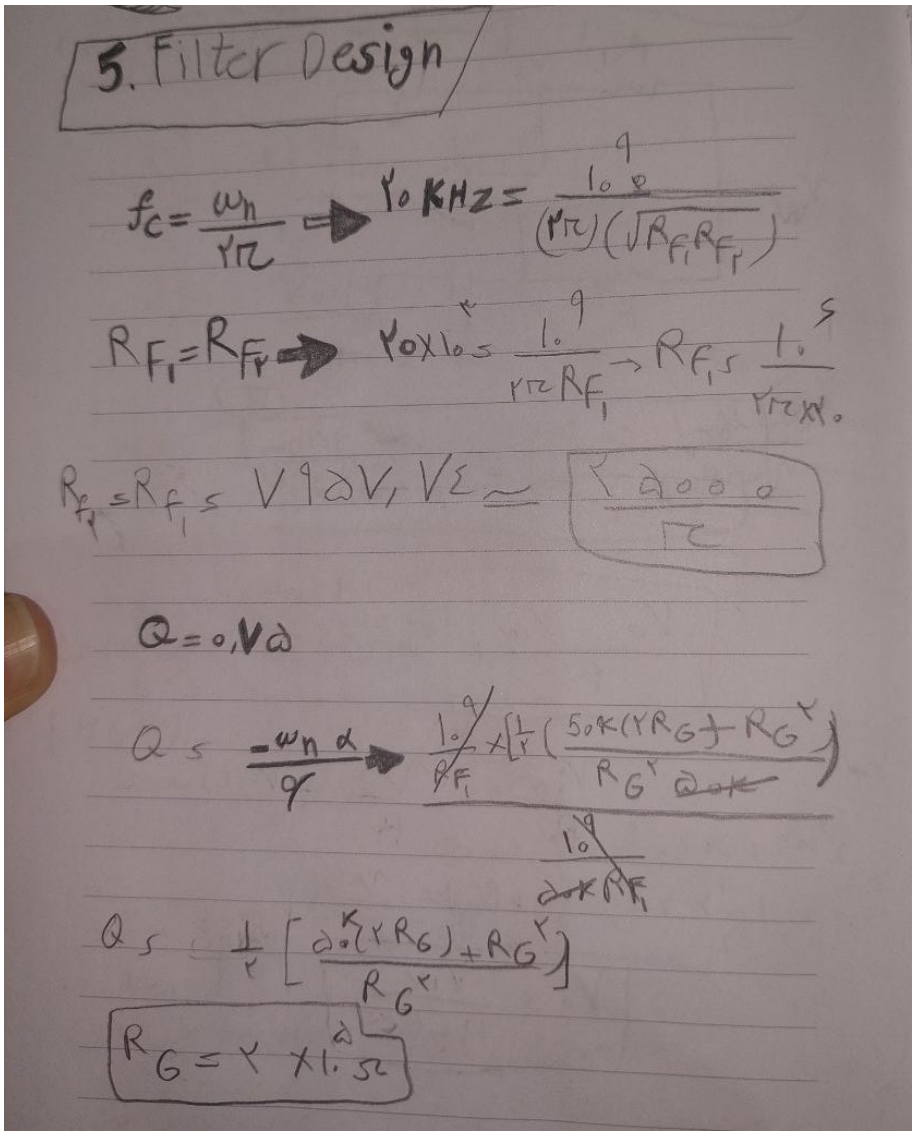


Figure 3: Handwritten solution for Part I

Problem 6: Simulation Study

Using **LTspice**, perform the following tasks:

- Draw the filter configuration.
- Run an AC analysis from 1 kHz to 100 kHz.
- Extract the cutoff frequency f_c, the quality factor Q, and the midband gain.
- Include the Bode magnitude plots.

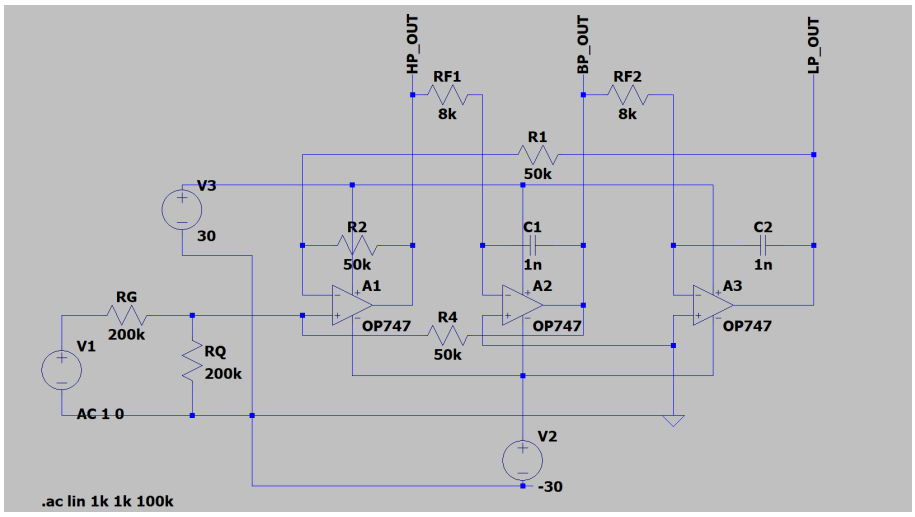


Figure 4: Handwritten solution for Part I



Figure 5: Handwritten solution for Part I

Problem 7: Comparison and Discussion

Compare your analytical and simulated results. Summarize the results in a table and calculate the percentage differences in the cutoff frequency f_c and the quality factor Q. Explain possible reasons for any observed discrepancies, including but not limited to:

- Finite op-amp gain and bandwidth limitations
- Numerical tolerances in simulation
- Sensitivity of the quality factor Q to resistor variations

Additionally, discuss how internal precision components contribute to performance stability, and explain why integrated active filter circuits are often preferred over discrete implementations.

Discussion and Comparison

Based on the analytical design, the resistor values were selected as $R_{F1} = R_{F2} = 7957.74\,\Omega$ and $R_G = 200\ k\Omega$, resulting in a quality factor of $Q = 0.75$. The assumption $R_{F1} = R_{F2}$

simplifies the circuit topology and leads to a symmetric filter structure, which improves design robustness and reduces sensitivity to component mismatch.

The obtained value of $Q = 0.75$ corresponds to a moderately damped response, providing a good trade-off between selectivity and stability. Such a value is less sensitive to resistor tolerances compared to higher-Q designs and is therefore well-suited for practical implementations.

In simulation, small deviations between analytical and simulated results may be observed. These discrepancies can be attributed to the finite gain and limited bandwidth of the operational amplifier, numerical tolerances in the simulation environment, and the inherent sensitivity of the quality factor Q to resistor variations.

Integrated active filter circuits typically offer improved matching between internal components, leading to enhanced performance stability. For this reason, they are often preferred over discrete implementations, where component tolerances and parasitic effects can significantly affect the filter response.

Bonus: Notch Filter Design Using SAF

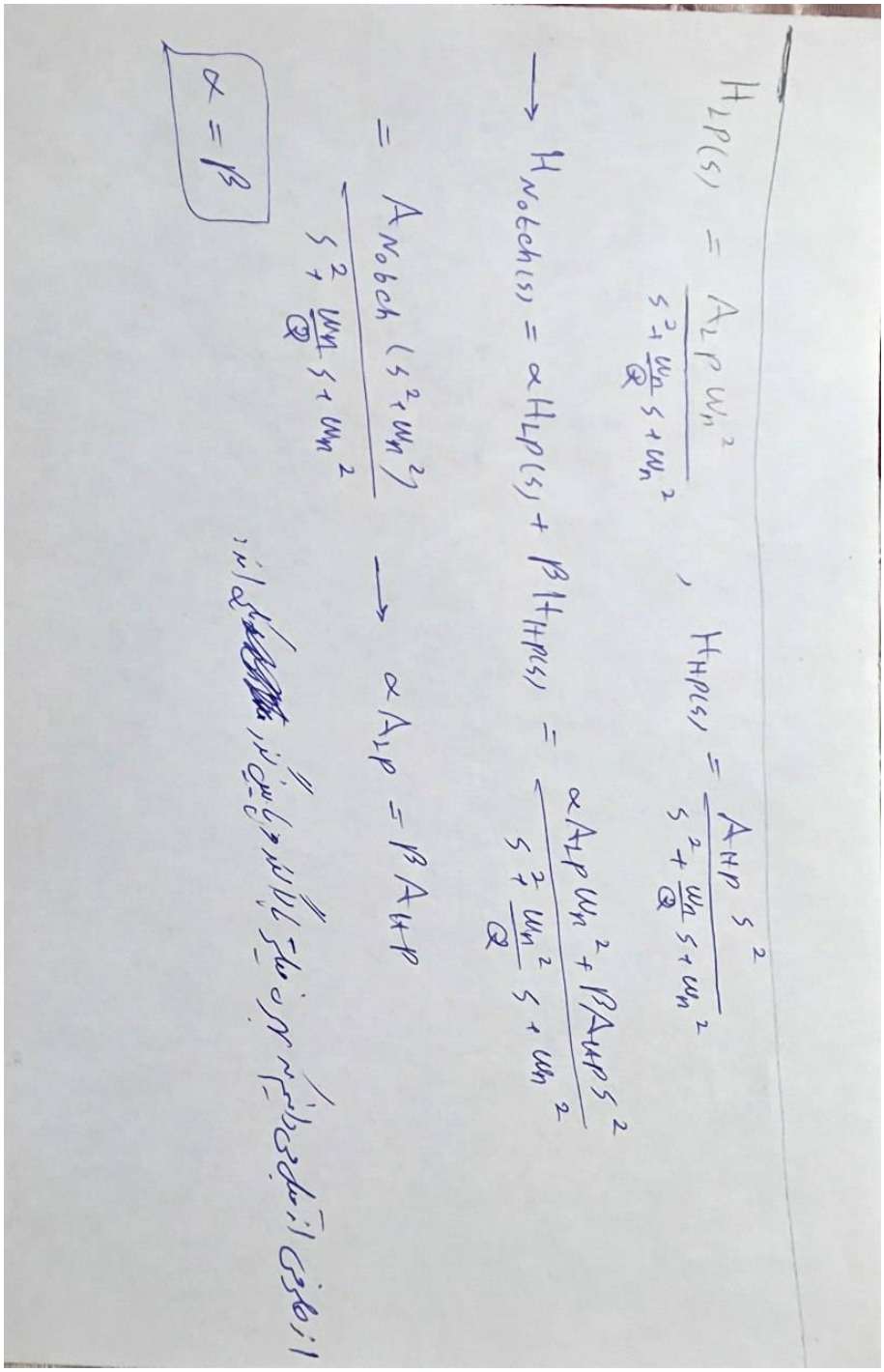


Figure 6: Handwritten solution for Part I

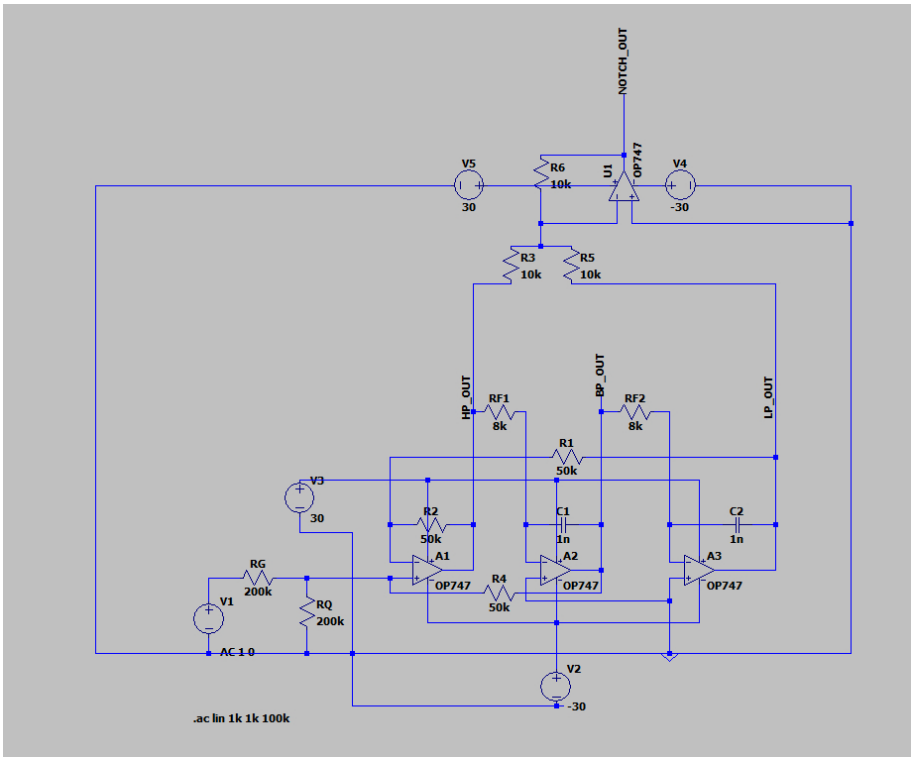


Figure 7: Handwritten solution for Part I

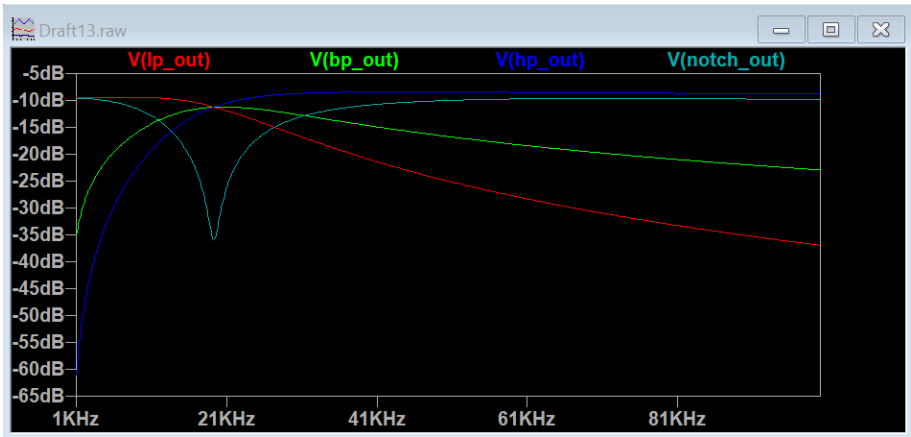


Figure 8: Handwritten solution for Part I

part two

Tasks

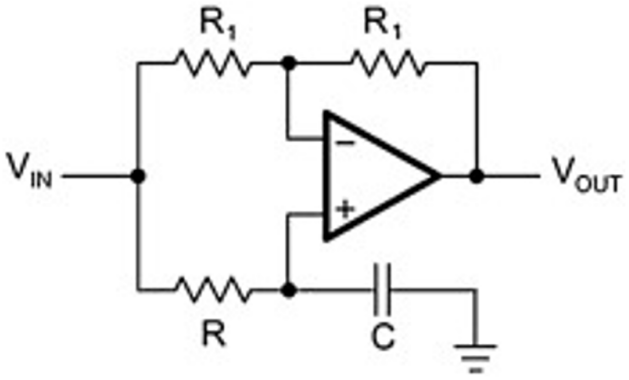


Figure 3: First-order active all-pass filter circuit

Figure 9: Handwritten solution for Part I

1. **Transfer Function:** Derive the transfer function,

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)},$$

   for the given circuit.

2. **Frequency Response Analysis:**

   - From the transfer function, determine the magnitude response,

$$|H(j\omega)|.$$

     Confirm that this circuit operates as an *all-pass* filter.

   - Derive the phase response,

$$\angle H(j\omega).$$

     Explain how this phase shift affects an input signal.

   - Define the relationship between the phase shift value and the circuit elements (e.g., resistor and capacitor values). Specify which circuit parameter controls the frequency at which a particular phase shift occurs.

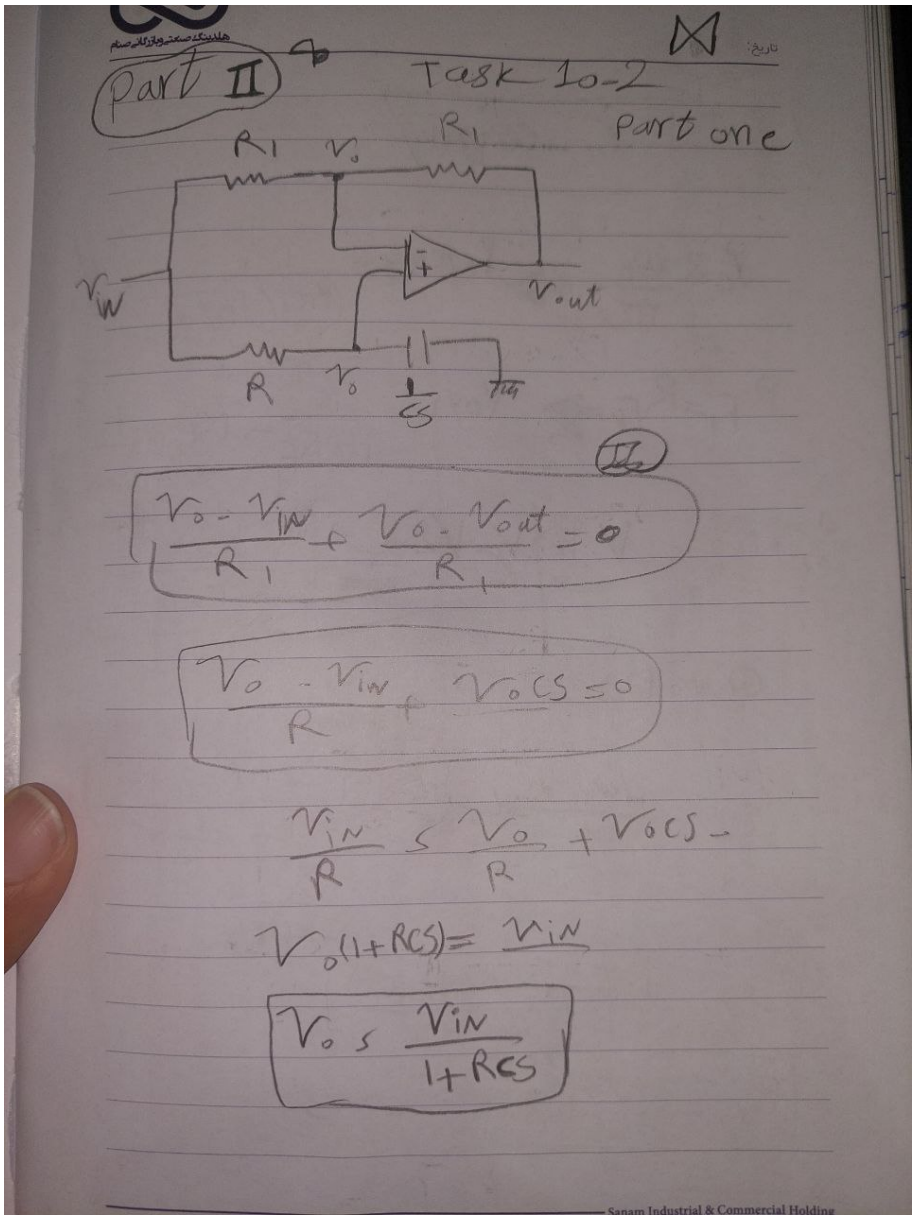


Figure 10: Handwritten solution for Part I

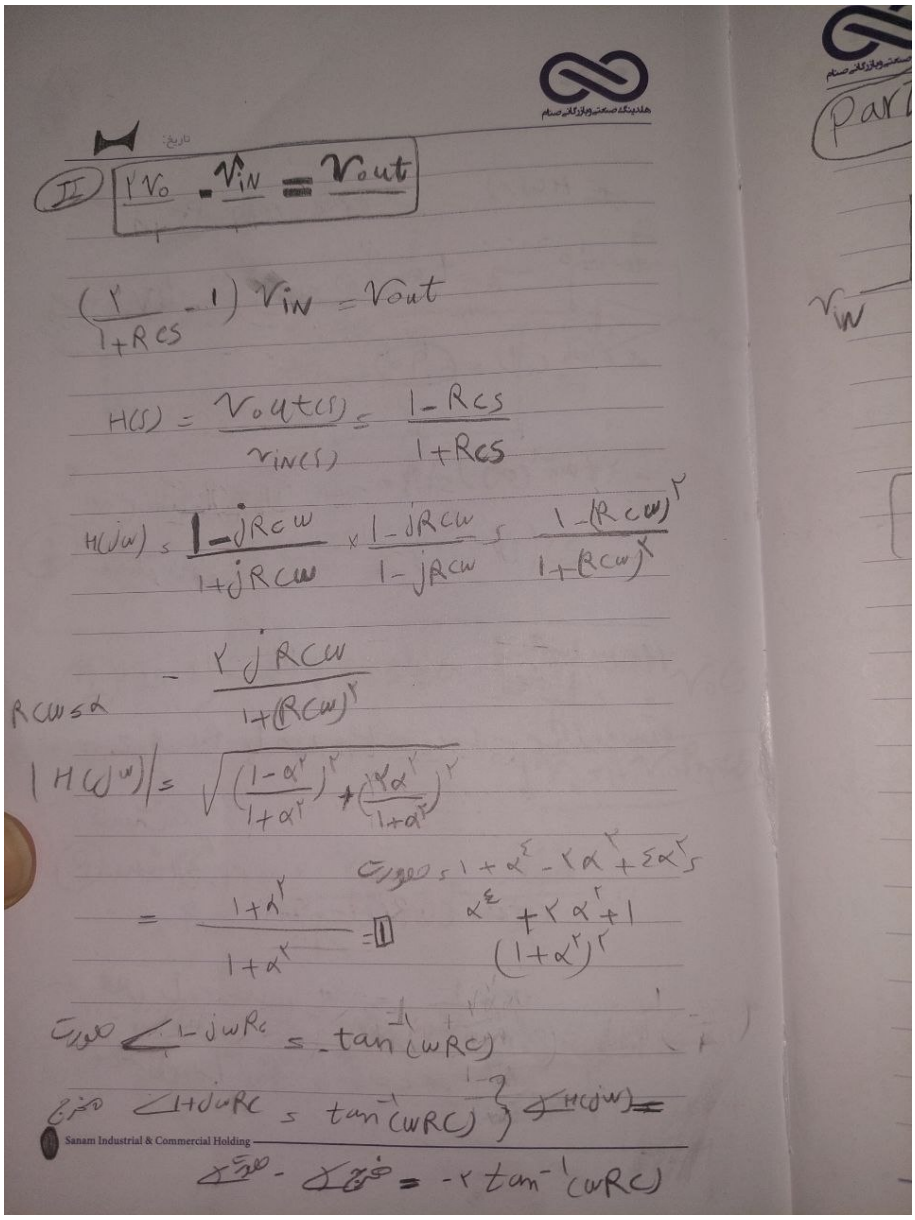


Figure 11: Handwritten solution for Part I

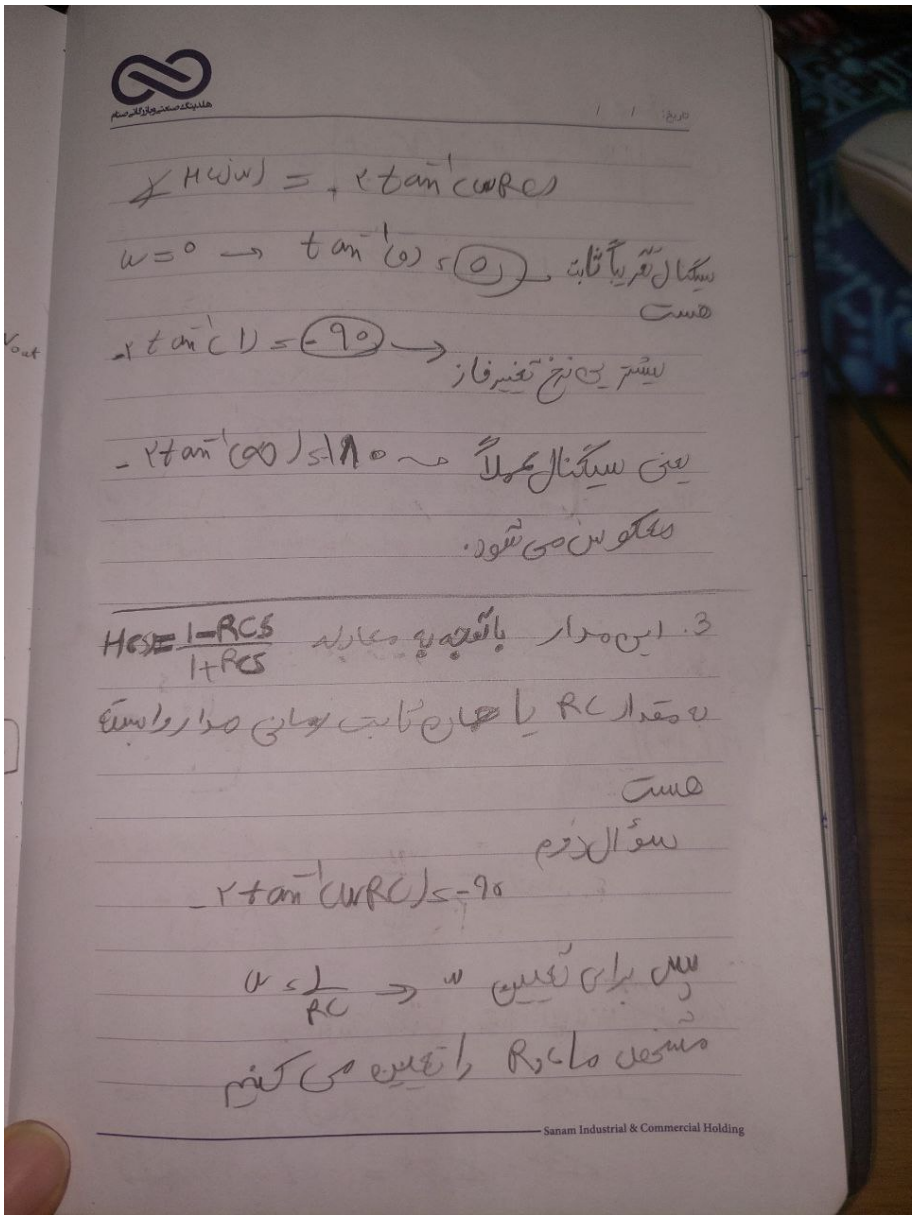


Figure 12: Handwritten solution for Part I

Part 2: Second-Order All-Pass Filter

The design can be extended to higher orders to obtain a more complex phase response. The circuit for a second-order active all-pass filter is shown in Figure 4.

Tasks

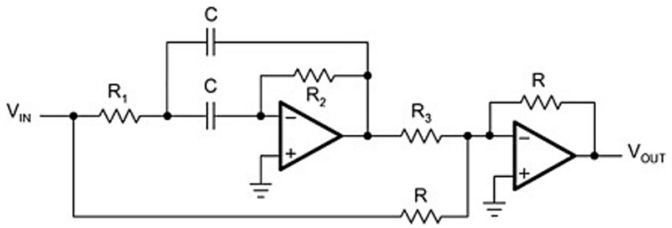


Figure 4: Second-order active all-pass filter circuit

Figure 13: Handwritten solution for Part I

1. **Transfer Function:** Find the transfer function $H(s)$ for this second-order circuit.
2. **Frequency Response Analysis:**
   - Determine the magnitude and phase response.
   - Explain how the phase shift effect differs from that of the first-order filter. How do the additional circuit elements provide more control over the phase response?
   - Define how the phase shift is determined by the values of the circuit components.

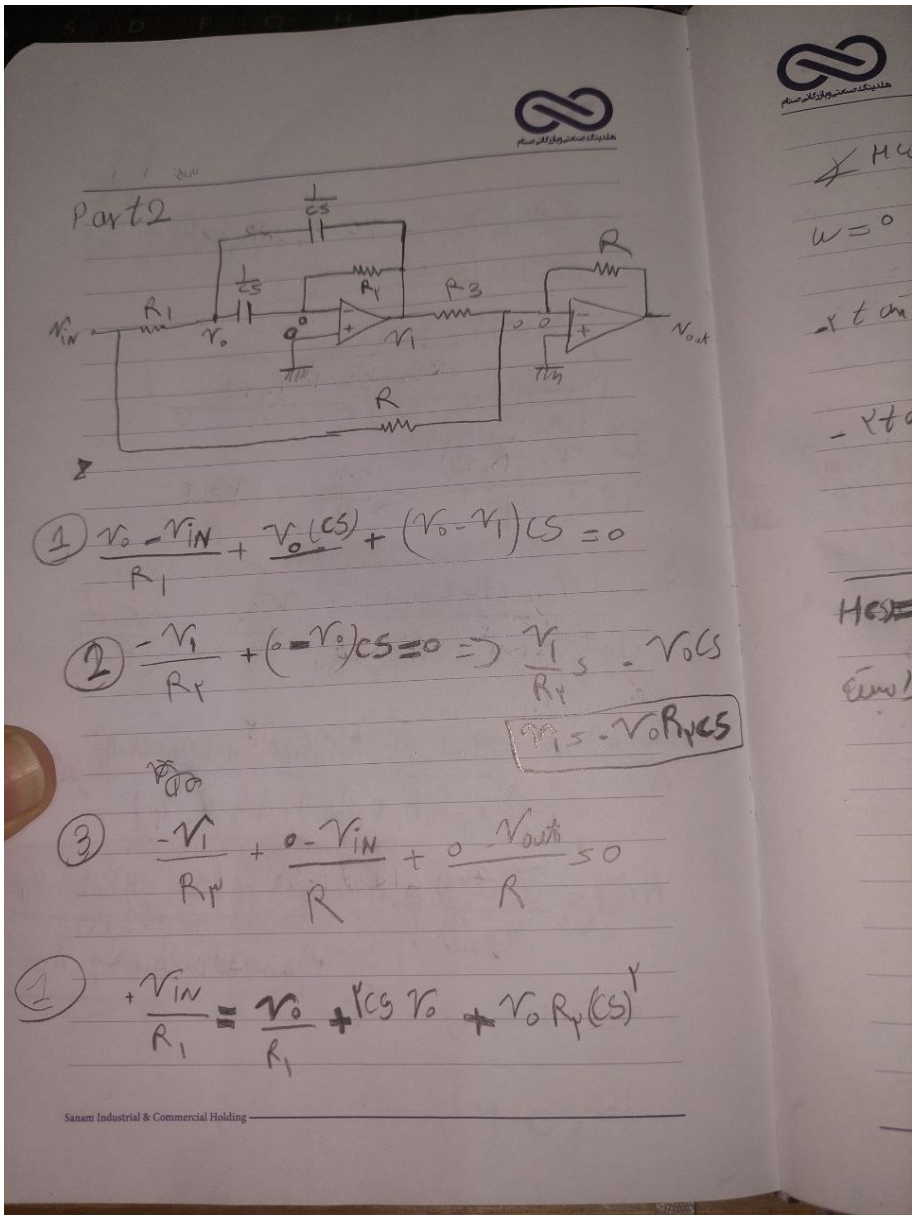


Figure 14: Handwritten solution for Part I

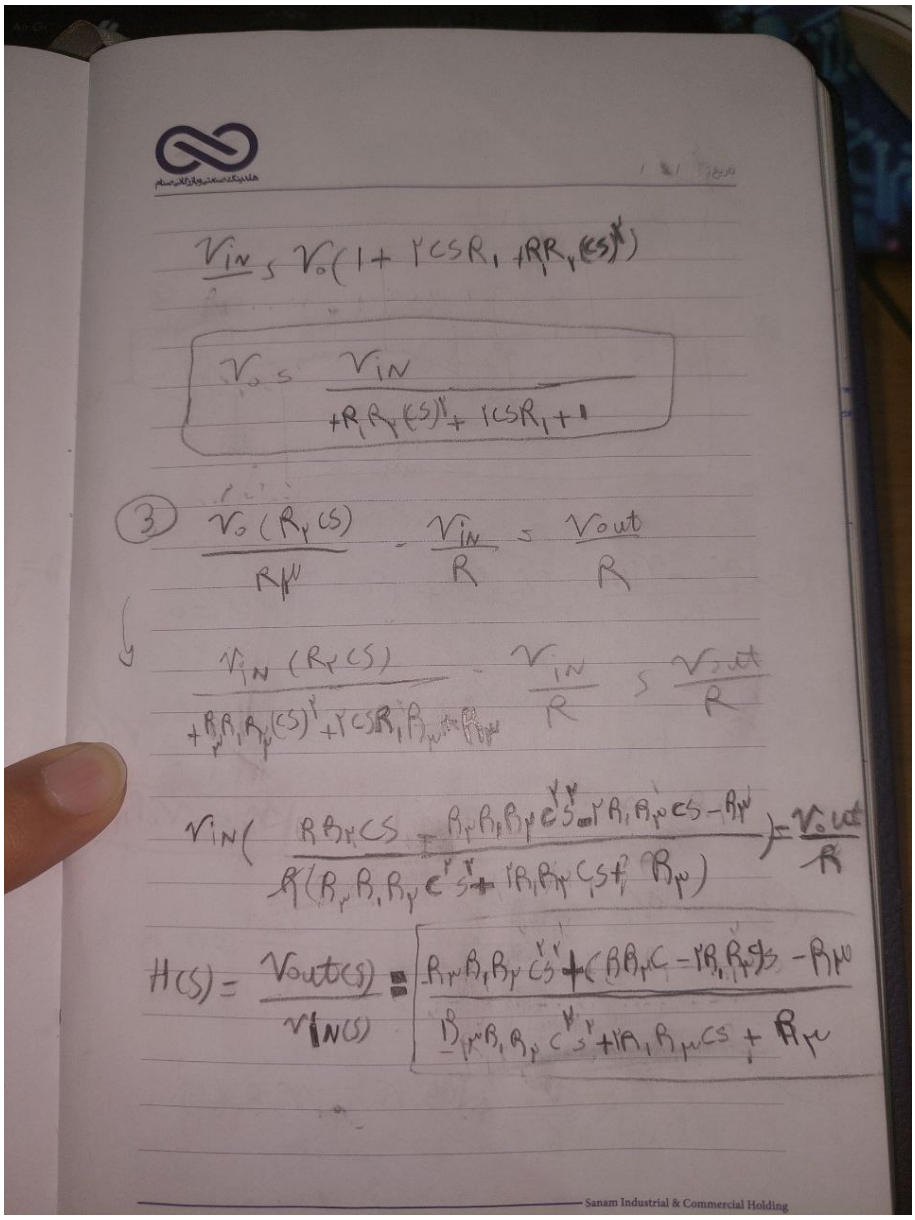


Figure 15: Handwritten solution for Part I

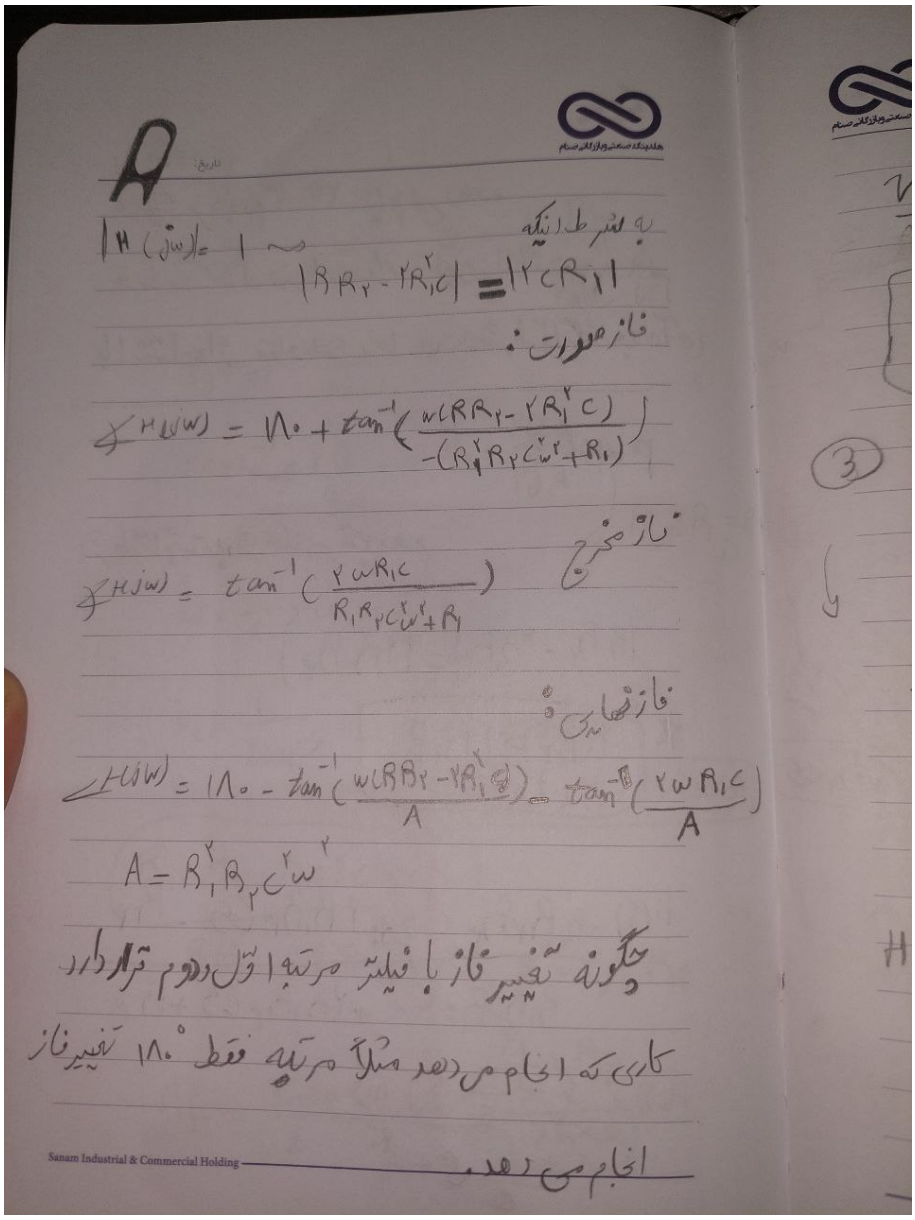


Figure 16: Handwritten solution for Part I

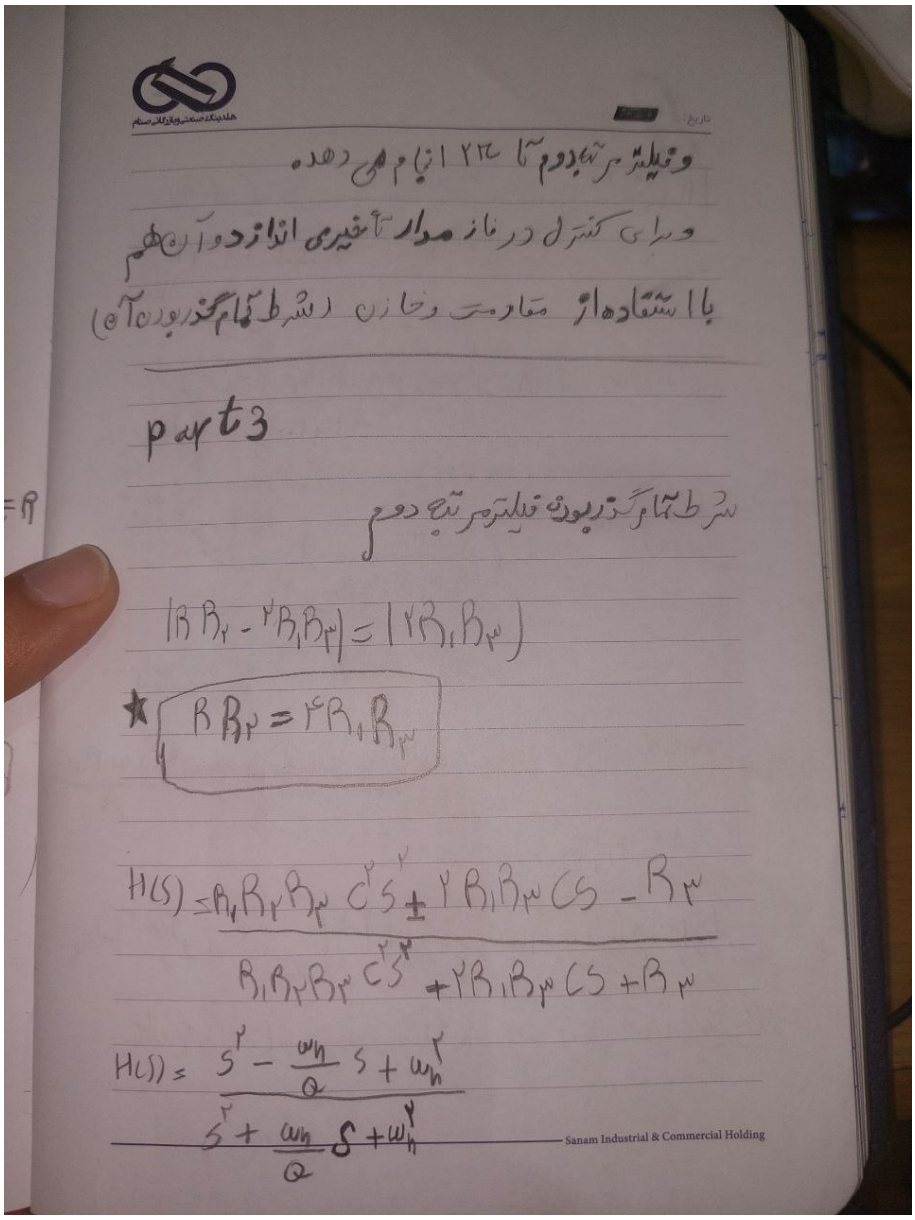


Figure 17: Handwritten solution for Part I

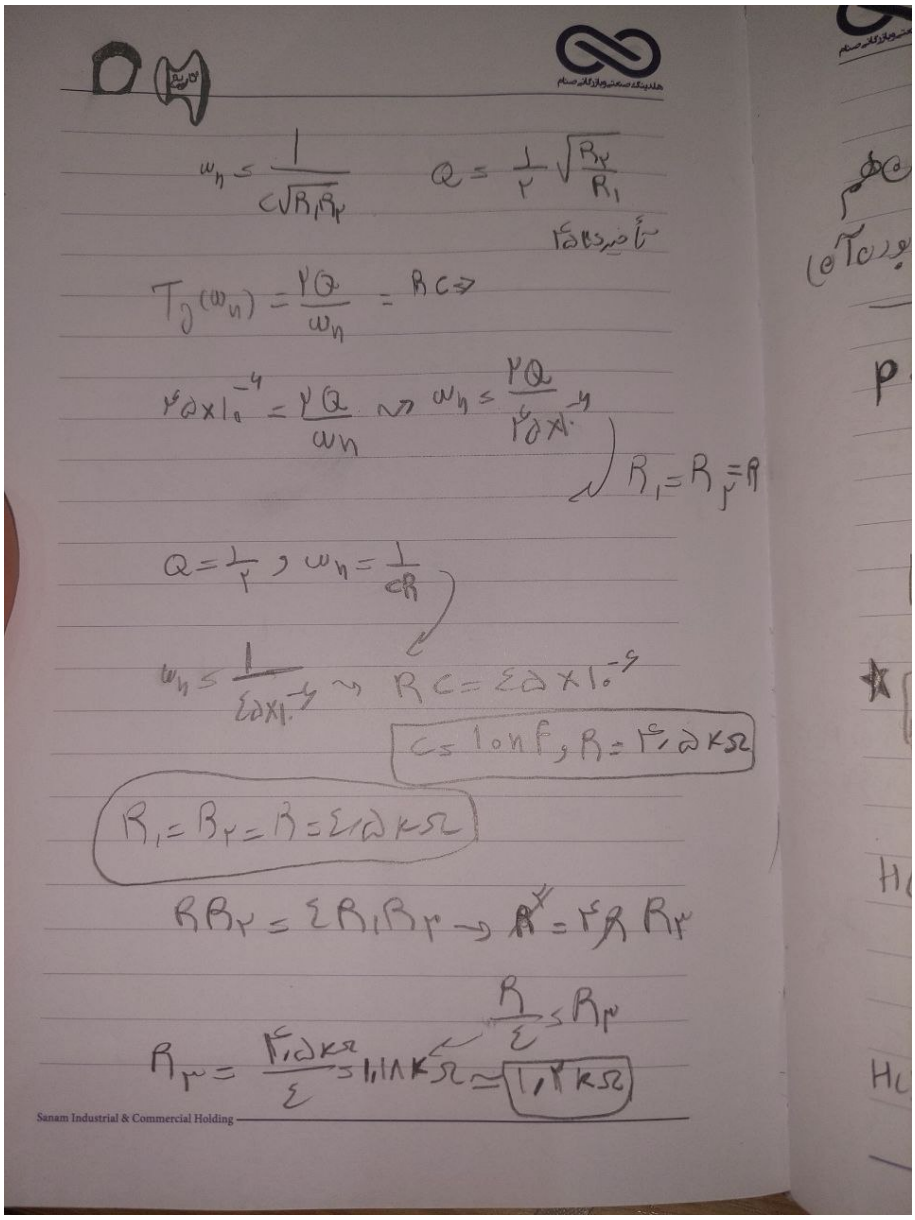


Figure 18: Handwritten solution for Part I

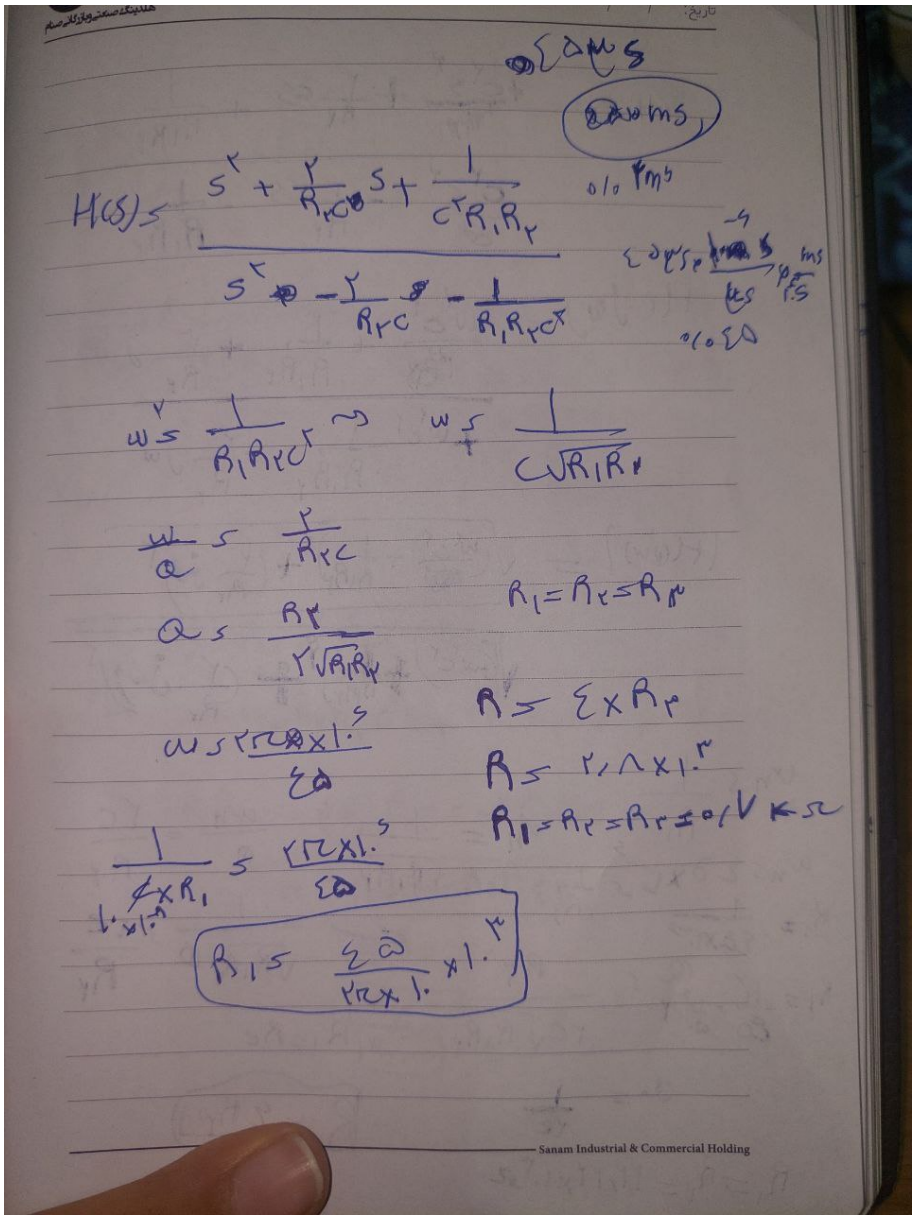


Figure 19: Handwritten solution for Part I

Part 3: Investigating Higher-Order Filters

The pattern of adding stages can be continued to create all-pass filters of any order. Figure 5 conceptually represents a seventh-order all-pass filter constructed from multiple first and/or second-order sections.

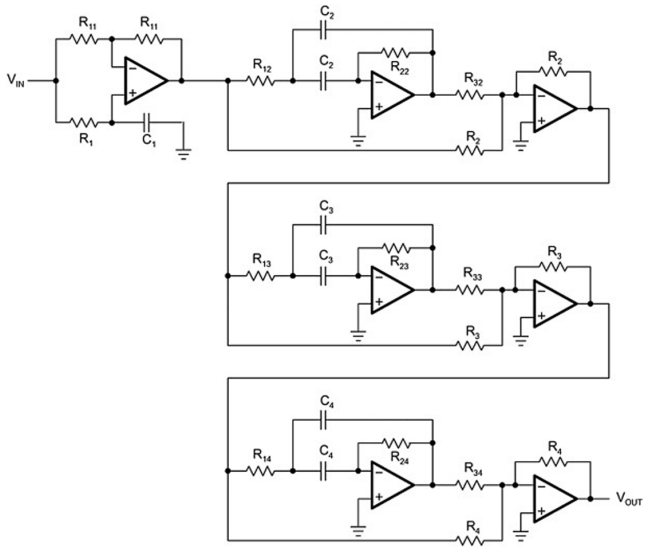


Figure 5: Block diagram of a seventh-order all-pass filter using cascaded stages

Figure 20: Handwritten solution for Part I

Question

Based on the patterns you observed in the first- and second-order filters, predict how the **impulse response** of the filter changes as the order increases. Consider the following points in your reasoning:

- What is the relationship between the order of a filter and the duration of its impulse response?
- How does increasing the number of stages affect the total phase shift a signal experiences?
- What would be the visual difference between the impulse response of a first-order and a tenth-order all-pass filter?

Effect of Filter Order on the Impulse Response

Answer to Question 1: Relationship between filter order and impulse response duration

The order of a filter is directly related to the number of poles in its transfer function and, consequently, to the number of energy storage elements in the system. As the filter order increases, the system exhibits a higher degree of memory, causing the impulse response to extend over a longer period of time. Therefore, higher-order filters produce impulse responses that are longer in duration and more complex compared to lower-order filters.

Answer to Question 2: Effect of increasing the number of stages on total phase shift

Each all-pass stage introduces a frequency-dependent phase shift while maintaining a unity magnitude response. When multiple stages are cascaded, the individual phase shifts add together, resulting in a larger cumulative phase shift. Consequently, increasing the number of stages significantly increases the total phase shift and the associated group delay experienced by the signal.

Answer to Question 3: Visual difference between first-order and tenth-order impulse responses

A first-order all-pass filter exhibits a simple impulse response characterized by a dominant peak and a rapid decay. In contrast, a tenth-order all-pass filter produces an impulse response that is spread over a much longer time interval and contains multiple oscillations. Visually, the higher-order impulse response appears more oscillatory and extended, reflecting the increased phase distortion and time dispersion introduced by the additional stages.

Design of an Analog Noise Cancelling Circuit Using an All-Pass Filter

Question

In practical Active Noise Control (ANC) systems, why is a delay component necessary between noise capture and cancellation?

Answer

In practical ANC systems, a delay component is necessary to ensure proper time alignment between the incoming noise and the generated anti-noise signal at the cancellation point. Since sound requires a finite amount of time to propagate from the noise source to the point of cancellation, and additional delays are introduced by signal processing and hardware components, the anti-noise signal must be delayed accordingly.

This delay allows the anti-noise signal to arrive with the correct phase (ideally 180° out of phase) relative to the noise, enabling effective destructive interference. Without proper delay compensation, the anti-noise may arrive too early or too late, resulting in incomplete noise cancellation or even noise amplification.

First Stage

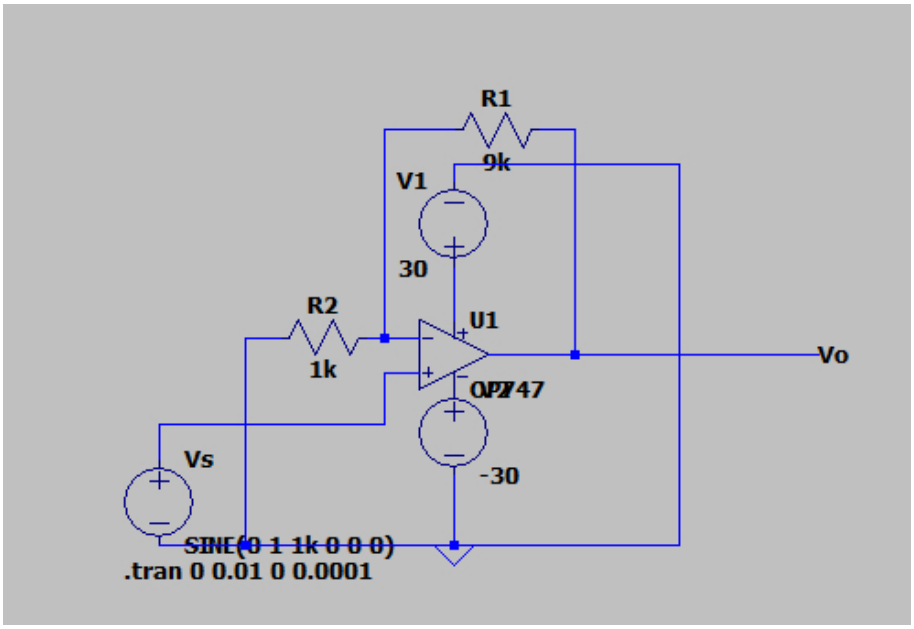


Figure 21: Handwritten solution for Part I

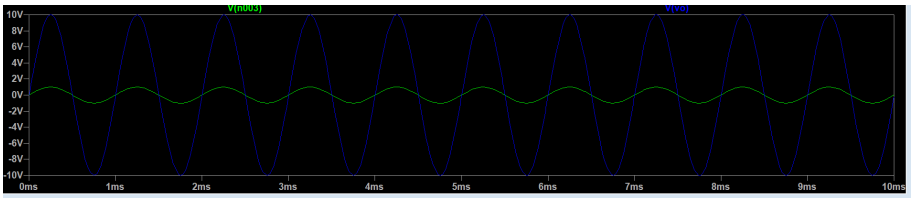


Figure 22: Handwritten solution for Part I

Second Stage

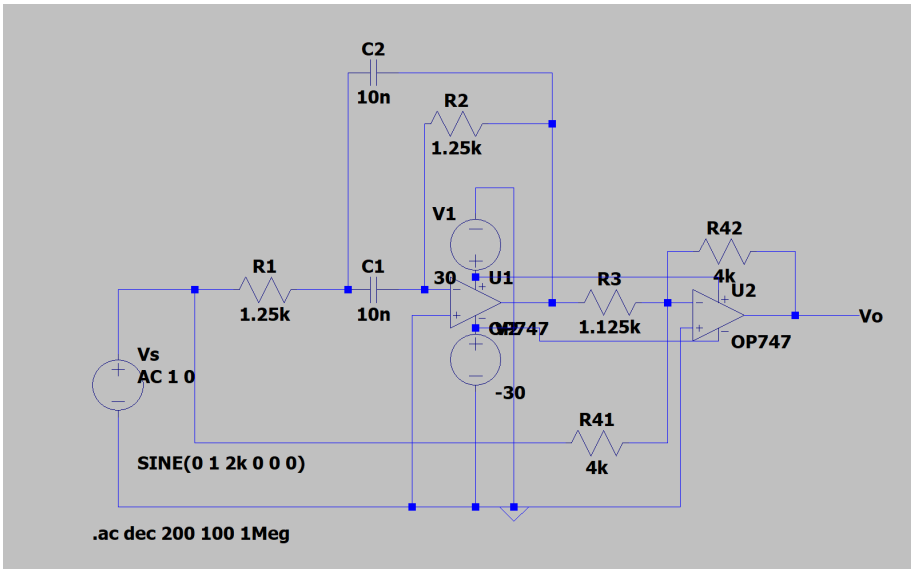


Figure 23: Handwritten solution for Part I

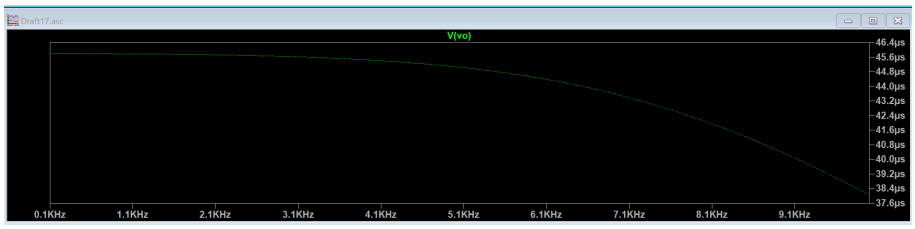


Figure 24: Handwritten solution for Part I

Third Stage

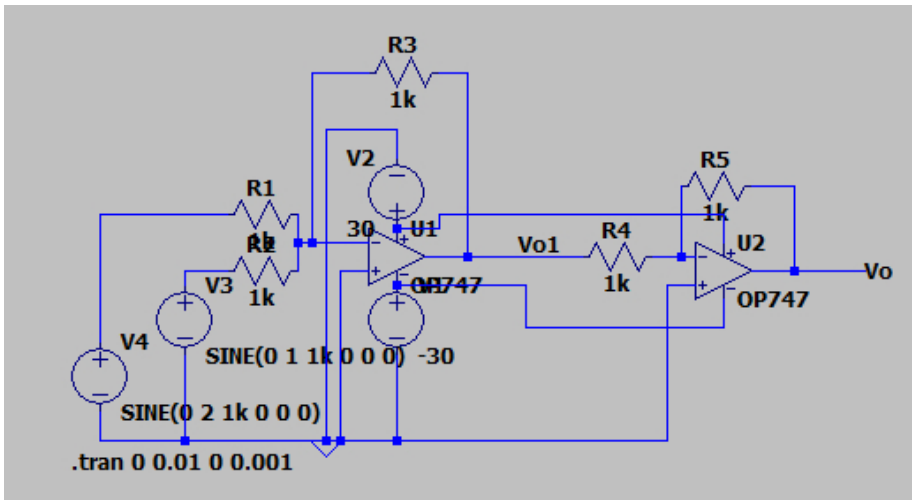


Figure 25: Handwritten solution for Part I

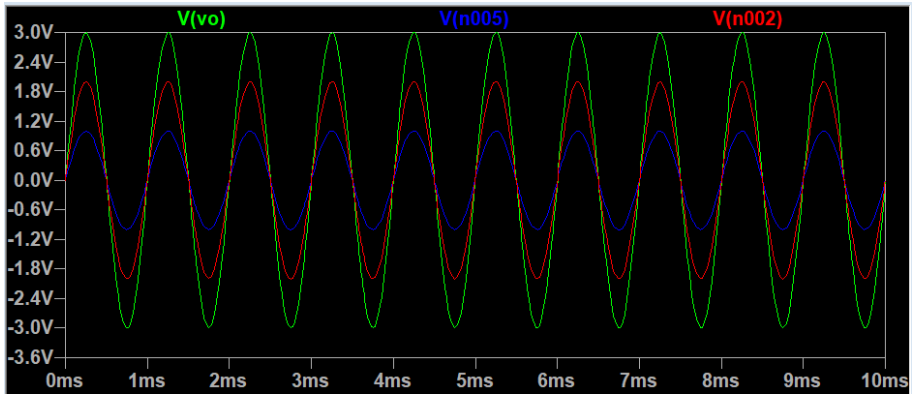


Figure 26: Handwritten solution for Part I

Overall System Description

This project implements an analog active noise cancellation (ANC) signal chain composed of several cascaded stages. The objective is to process a noise reference signal, apply appropriate gain and delay, and combine it with a noisy music signal in order to reduce the noise content at the output.

System Operation Stages

Stage 1: Noise Injection and Frequency Analysis

At the first stage, only the noise signal is applied as the input to the circuit. A conversion element is used to properly inject the noise signal into the analog domain. The noise is then analyzed in the frequency domain inside the circuit to identify its spectral characteristics.

Stage 2: Noise Amplification

The noise signal is passed through a gain stage with a gain factor of 10. This amplification ensures that the noise signal has sufficient amplitude for effective processing in the subsequent stages.

Stage 3: Time Delay Using a Second-Order All-Pass Filter

A time delay of approximately 45 μs is introduced using a second-order all-pass filter. This filter modifies the phase response of the noise signal while maintaining a constant magnitude response.

Stage 4: Signal Summation Without a Subtractor

Due to the symmetric (phase-inverting) nature of the second-order all-pass filter, an explicit subtraction block is not required. The delayed noise signal is directly summed with the music signal that already contains noise, enabling destructive interference.

Stage 5: Output Signal Inversion

An additional inverting stage is used to correct the signal polarity and ensure that the final output signal has the correct orientation.

Stage 6: Audio Reconstruction and Output

The processed signal is converted back into an audio waveform and exported as a `.wav` file. This output is used for listening and evaluation of the noise cancellation performance.

part three

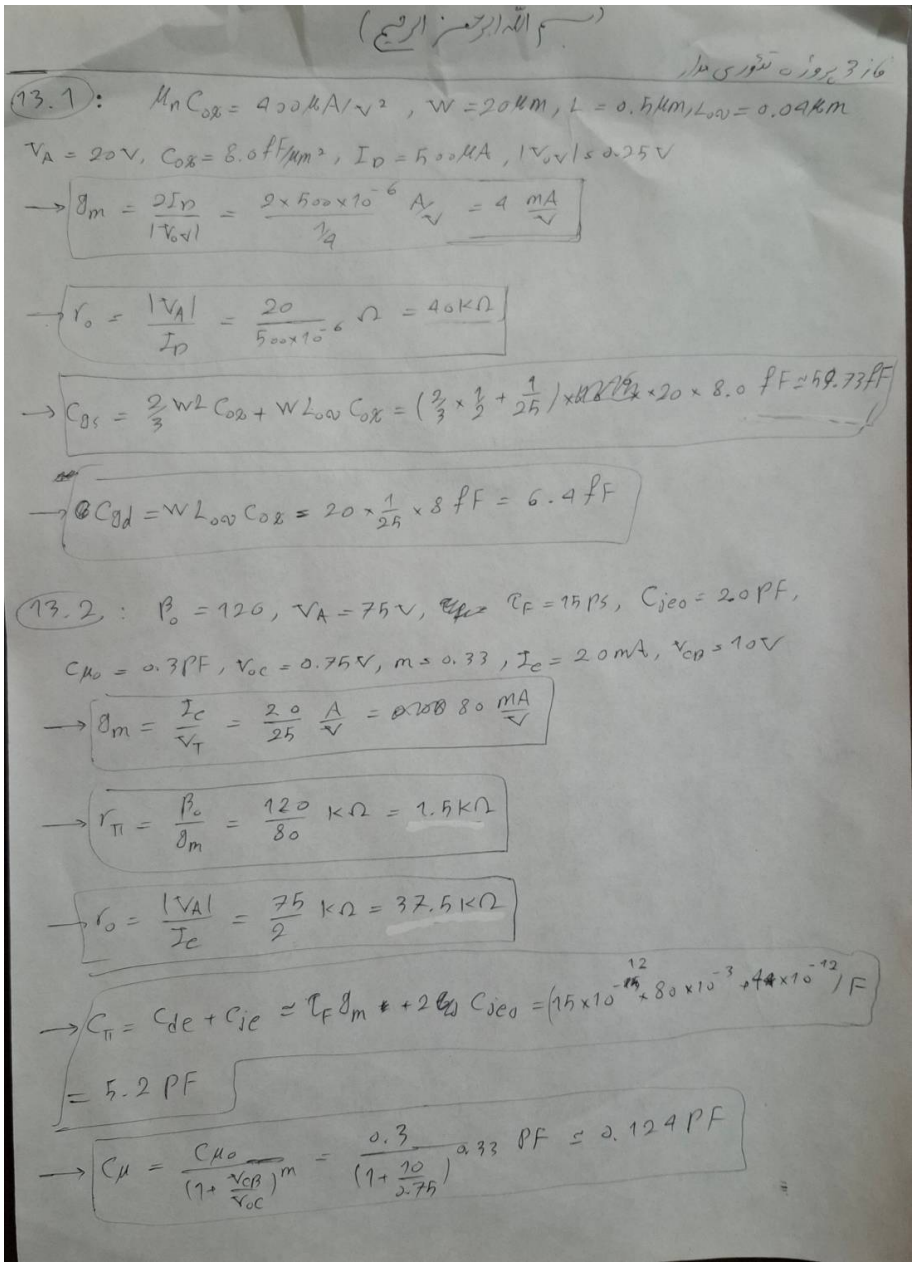


Figure 27: Handwritten solution for Part I

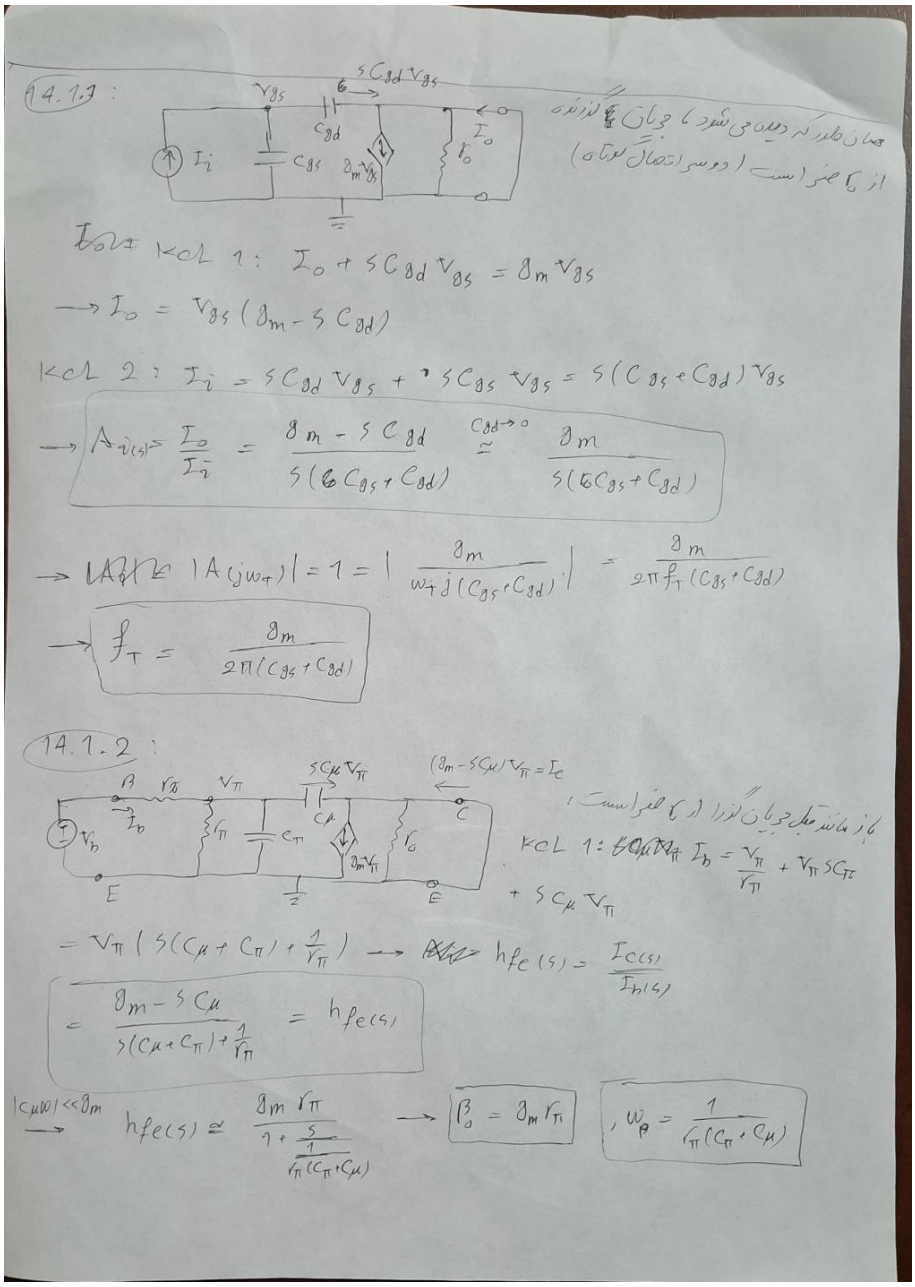


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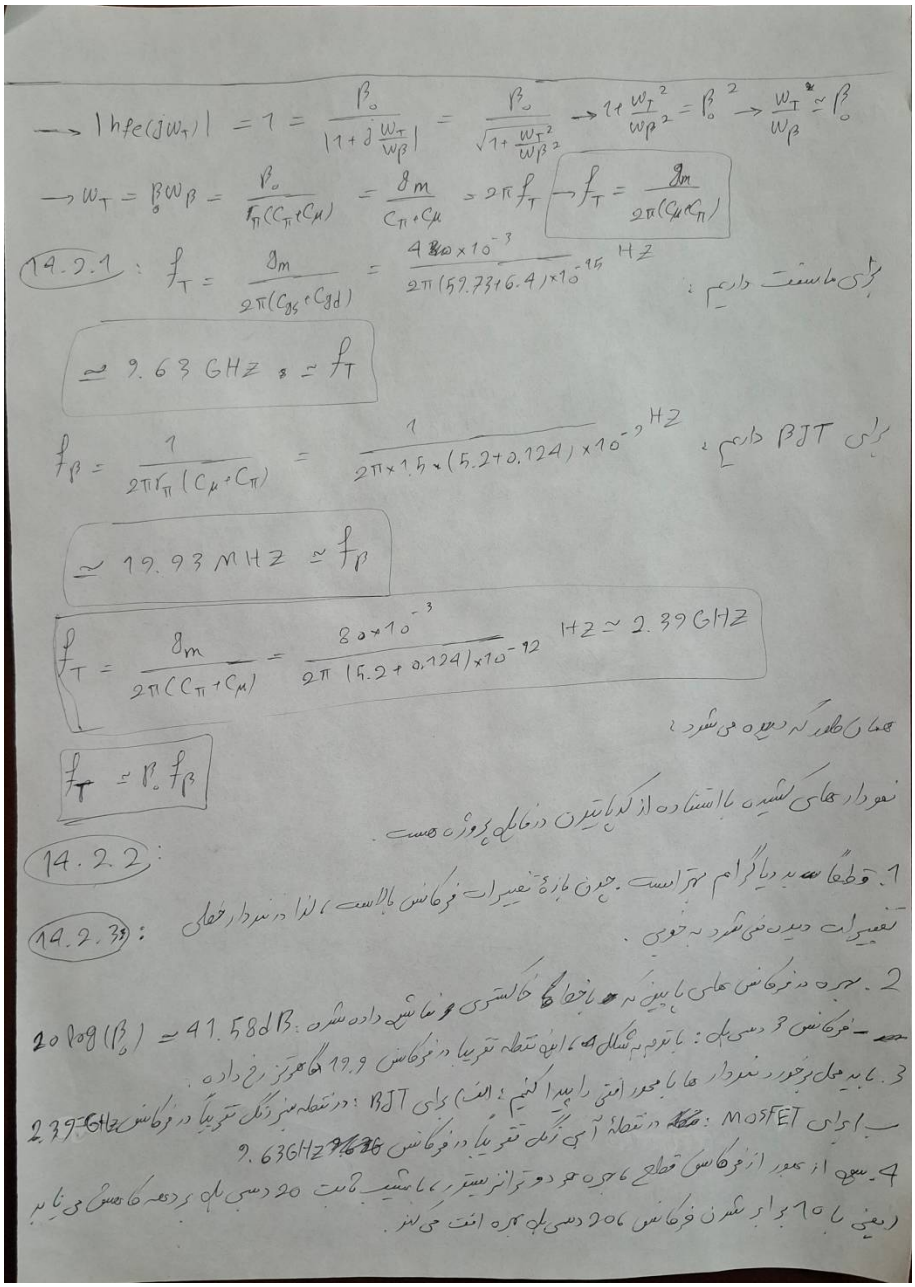


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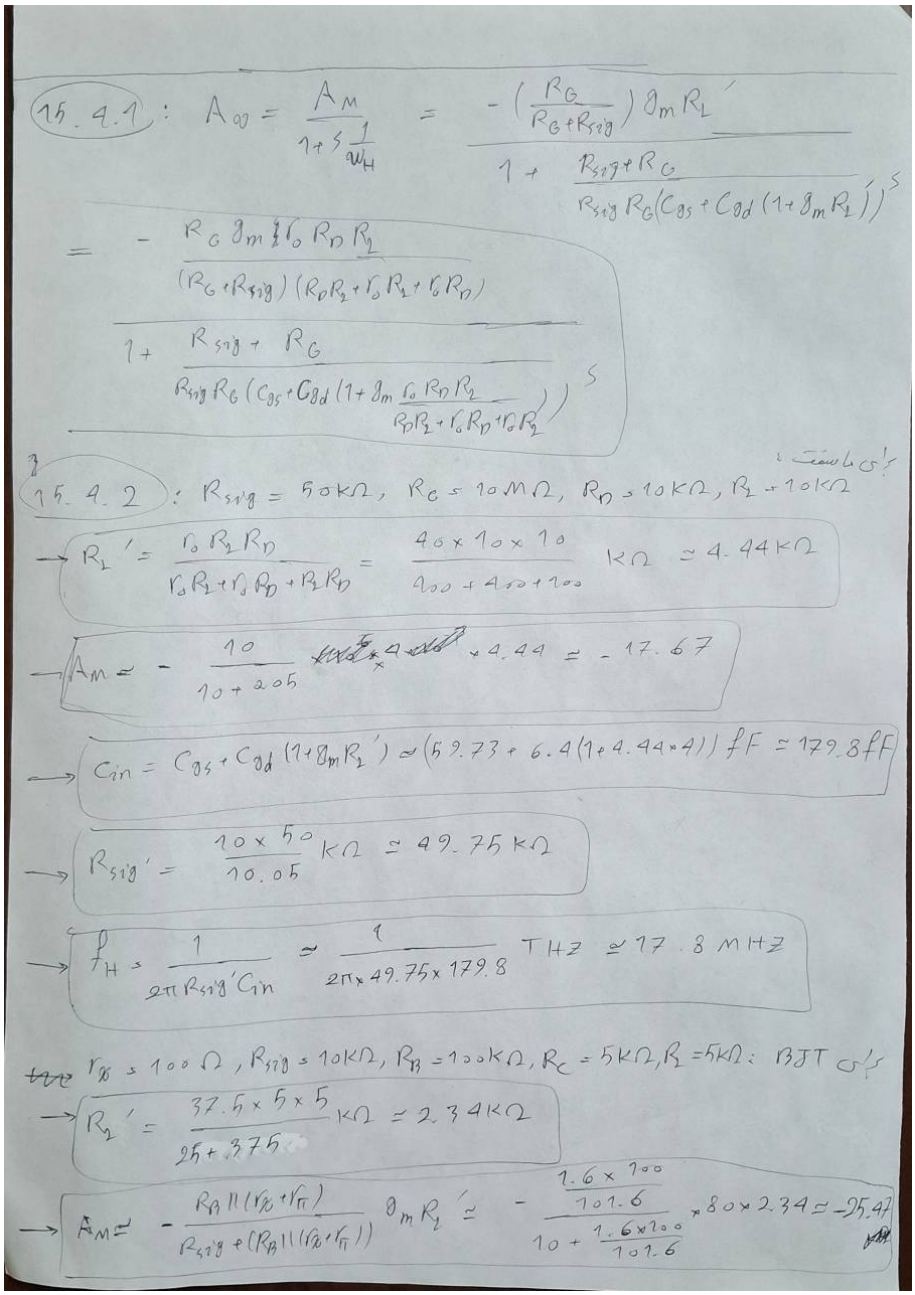


Figure 30: Handwritten solution for Part I

Figure 31: Handwritten solution for Part I

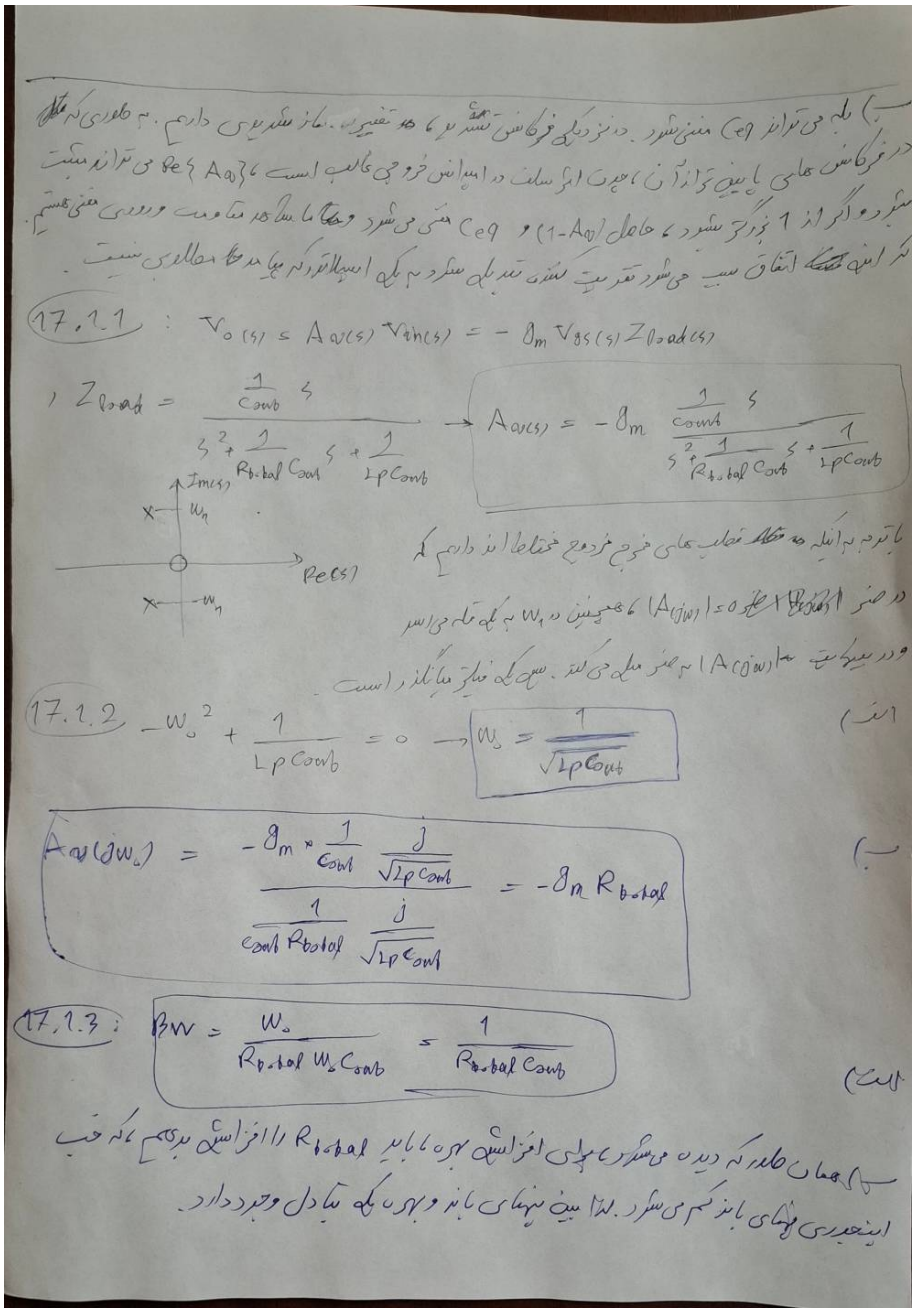


Figure 32: Handwritten solution for Part I